

AIEBOK Companion

Offline study companion for the AI Engineering Body of Knowledge. The full interactive site includes labs, notebooks, and search. This PDF summarizes structure, books, and a glossary sample.

Guided books

01. Foundations of Intelligence - Understand intelligence as search, learning, representation, decision-making, and feedback before studying language models.
02. Machine Learning Systems - Learn how models are trained, validated, diagnosed, and integrated into reliable software systems.
03. Language and Representation - Understand how language becomes computable representations and why embeddings enable semantic systems.
04. Transformers and Foundation Models - Understand the architecture, training, inference, and model families behind modern generative AI.
05. Prompt and Context Engineering - Design the information, instructions, state, and output boundaries that make model behavior useful and testable.
06. Knowledge and Retrieval Systems - Build grounded knowledge systems whose ingestion, retrieval, generation, and citations can be evaluated separately.
07. Reasoning and Tool Use - Understand when inference-time search, planning, verification, and external tools improve task outcomes.
08. Agent Systems - Design agentic systems as bounded stateful architectures rather than treating autonomy as a model feature.
09. AI Software and Product Engineering - Use specifications, evaluations, secure development practices, and product discovery to deliver useful AI features.
10. Evaluation, Safety, and Governance - Build evidence that AI systems meet task, safety, security, compliance, and business requirements.
11. Training, Serving, and AI Operations - Understand model adaptation, efficient inference, deployment, observability, release, and lifecycle operations.
12. Cloud and Enterprise AI Architecture - Design secure, governed, resilient AI platforms and map them to managed services without losing vendor-neutral reasoning.
13. Multimodal and Frontier Systems - Understand multimodal pipelines and evaluate emerging AI capabilities through enduring principles and evidence.

Glossary sample (A-C)

a: A* expands the lowest estimated total-cost node first, combining path cost $g(n)$ with heuristic $h(n)$ toward the goal.

a-b-testing: A/B testing compares product variants on live users with ethical guardrails and pre-registered metrics.

a-b-tests: A/B tests compare prompt or context variants on live traffic with guardrail metrics.

ablations: Ablations remove components to measure contribution - essential for judging which mechanism drives reported ...

abstention: Abstention lets a system refuse or defer when confidence is insufficient, routing cases to humans or safe...

acceptance-criteria: Acceptance criteria are pass/fail conditions for feature completion - testable, unambiguous, tied ...

accessibility: Accessibility ensures AI features work with screen readers, keyboard navigation, and assistive tech - no...

action-spaces: Action spaces define allowed agent operations - click, type, scroll, API call - with granularity affecting...

activations: Activation functions introduce nonlinearity - ReLU, GELU, sigmoid - without which deep networks collapse to ...

adaptive-rag: Adaptive RAG chooses retrieval depth, query rewrite, or no retrieval based on question type and confide...

adoption: Adoption tracks who uses the feature, how often, and whether usage persists after novelty fades.

adversarial-tests: Adversarial tests probe injection, jailbreaks, edge inputs, and abuse scenarios.

agency: Agency is goal-directed action selection in a loop - observe, decide, act - rather than a single model call.

ai-coding-agents: AI coding agents autonomously edit repositories given goals, tools, and constraints.

ai-gateways: AI gateways centralize model access with auth, rate limits, logging, routing, and policy enforcement for...

ai-inventory: AI inventory catalogs models, datasets, prompts, and features with owners, risk tier, and dependencies.

aks-and-functions: AKS and Azure Functions run containerized model servers and event-driven AI glue code on Azure.

amazon-bedrock: Amazon Bedrock provides managed access to foundation models from multiple providers via unified AWS A...

ambiguity: Ambiguity arises when the same text supports multiple interpretations without disambiguating context.

ann-indexes: Approximate nearest neighbor indexes - HNSW, IVF, LSH - trade recall for speed at million-plus scale.

answer-validation: Answer validation runs programmatic checks - schema, arithmetic, citation alignment - on model outputs...

approval-gates: Approval gates pause execution until authorized humans confirm high-impact actions.

attention-masks: Attention masks zero out disallowed positions - future tokens in decoding, padding, or cross-segment b...

audit: Audit logs record who accessed which models, documents, and tools with immutable retention for compliance.

audit-evidence: Audit evidence collects eval reports, approvals, change logs, and incident records demonstrating cont...

authentication: Authentication verifies identity of users, clients, and services before access to models, tools, or d...

authorization: Authorization ensures retrieved and acted-upon data respects user permissions - not just authentication.

autoencoders: Autoencoders learn compressed representations by reconstructing inputs through a bottleneck layer.

autonomy: Autonomy is how much discretion the system has to choose actions without human approval.

autoscaling: Autoscaling adjusts inference replica count based on CPU, GPU utilization, or queue depth.

azure-ai-foundry: Azure AI Foundry is Microsoft's unified portal for model deployment, fine-tuning, evaluation, and a...

azure-ai-search: Azure AI Search provides hybrid lexical-vector search, semantic ranker, and skill pipelines for RAG ...

azure-openai: Azure OpenAI Service hosts OpenAI models in Azure regions with private networking, content filters, and...

backpropagation: Backpropagation applies the chain rule to compute gradients through layered computations efficiently.

backtracking: Backtracking abandons partial solutions that fail constraints and returns to earlier choices.

baseline-workflow: Baseline workflow documents how users solve the task today - time, errors, tools - before AI intervent...

baselines: Baselines are simple reference methods - majority class, linear model, keyword rules - that quantify what comp...

batching: Batching groups requests to amortize GPU kernel overhead, improving throughput at possible latency cost.

behavior-versus-knowledge: Behavior changes how the model acts - tone, format, policy - while knowledge is factual content.

benchmarks: Benchmarks standardize task comparisons - MMLU, HumanEval, BEIR - but may not reflect your production distrib...

best-of-n: Best-of-N generates N candidates and selects the best by a scorer or verifier.

bias-and-variance: Bias is systematic underfitting from overly simple models; variance is sensitivity to training noi...

bm25: BM25 ranks documents by weighted term frequency with length normalization and term-frequency saturation.

bottlenecks: Information bottlenecks force compressive representations - fixed-size context vectors or limited bandwidth...

bounded-rationality: Bounded rationality acknowledges limited compute, time, memory, and information - systems must sat...

bpe: Byte-pair encoding iteratively merges frequent symbol pairs to build a subword vocabulary from corpus statistics.

breadth-first-search: Breadth-first search expands nodes level by level, guaranteeing shortest path in unweighted gra...

budgets: Budgets cap tokens, tool calls, wall time, or dollars per task or session.

build-versus-buy: Build versus buy weighs custom AI development against vendor APIs and platforms on control, cost, a...

caching: Caching stores prompt prefixes, embeddings, or completions to cut latency and cost.

calibration: Calibration means predicted probabilities align with observed frequencies - 70% confidence should be right...

canaries: Canaries route small traffic percentage to new versions before full rollout.

cancellation: Cancellation stops in-flight agent work cleanly - revoke leases, abort tool calls, compensate partial eff...

capability-decomposition: Capability decomposition splits intelligence into perception, memory, learning, planning, a...

checkpoints: Checkpoints persist durable agent state so interrupted runs resume without repeating side effects.

chunking: Chunking splits documents into index units sized for retrieval precision and generation context.

citation-precision: Citation precision measures whether cited sources actually support the adjacent claims.

citations: Citations link UI claims to source passages users can verify.

classification: Classification assigns inputs to discrete categories via scores converted to labels.

cloud-iam: Google Cloud IAM binds roles to identities for least-privilege access to Vertex, Storage, and BigQuery in ...

cloud-run-and-gke: Cloud Run and GKE deploy serverless containers and Kubernetes GPU workloads on Google Cloud.

cloudwatch-and-iam: CloudWatch and IAM deliver AWS monitoring, alerting, and access control for AI workloads - metrics,...

clustering: Clustering groups unlabeled points by similarity - k-means, hierarchical, or density methods.

code-review: Code review evaluates correctness, security, and maintainability of changes - including agent-written code.

compensation: Compensation undo or offsets partial effects when later steps fail - Saga pattern for agents.

component-evals: Component evals test retrieval, generation, tools, and UX stages independently before end-to-end runs.

compression: Context compression summarizes, extracts, or prunes evidence to fit token limits while preserving decisi...

computer-use: Computer use agents perceive screens and emit mouse/keyboard actions to complete software tasks.

conditioning: Conditioning steers generation with extra inputs - text, masks, ControlNet edges, brand assets.

confidence-intervals: Confidence intervals quantify uncertainty in metric estimates from finite eval sets.

confusion-matrix: A confusion matrix counts predicted versus actual classes, exposing which errors dominate.

consensus: Consensus protocols align multiple agents on a decision before action - voting, debate, or judge model.

constraints: Constraints specify forbidden actions, length limits, formats, and scope boundaries in prompts.

containers: Containers package model servers with dependencies for reproducible deployment across environments.

contamination: Contamination occurs when eval examples leak into training data, inflating benchmark scores.

context-assembly: Context assembly is the pipeline that gathers instructions, state, evidence, tools, and examples in...

context-files: Context files - .

context-packing: Context packing fits selected passages into the token window respecting priority, citation needs, an...

context-poisoning: Context poisoning inserts false or misleading evidence into retrieval or memory stores to manipula...

context-traces: Context traces log the assembled prompt sections, token counts, and sources for debugging and complia...

Using this with GitHub Pages

Host: deploy MkDocs to GitHub Pages (static HTML). Labs: run Python locally or in GitHub Codespaces / Dev Container. Starter lab notebooks: open lab.ipynb from the labs/ directory in the repository. Print full site: use the Print Site plugin page in the Reference section.